Efficient solution of symmetry-unrestricted density functional theory using a conjugate gradient method

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1 OVERVIEW

Deformed, weakly-bound and superheavy nuclei still represent a challenge for **theoretical** models.

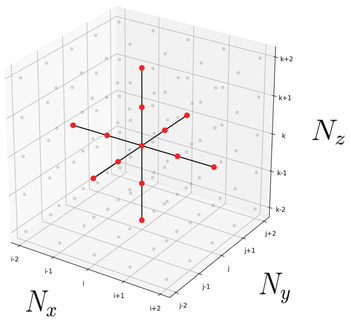
Energy density **functionals** can be used to make predictions, however numerical challenges still exist1.

Cartesian meshes have been used in recent years in an effort to reduce the numerical error of **basis** methods in deformed, drip line and superheavy systems.

Problem: Current methods are limited by slow **convergence** of the large eigenvalue problem and the need of human trial and error for optimal **parameter choice**.

$$N_{\text{basis}} = 2 \cdot N_x \cdot N_y \cdot N_z \sim 10^5$$

Solution: We effectively address these problems using the **Generalized Conjugate Gradient**[2].



2 METHODS & NUMERICAL ASPECTS

The core of the problem is the repeated solution of the **Schrödinger**'s equation on a **Cartesian** mesh:

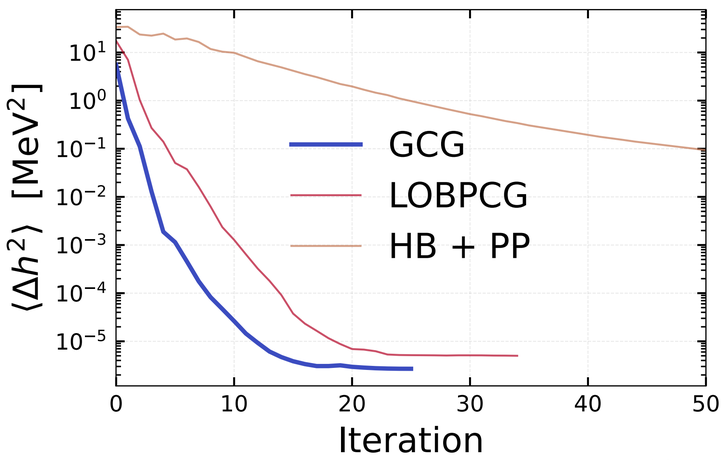
$$h(\boldsymbol{r}_i, \boldsymbol{r}_j)\psi_k(\boldsymbol{r}_j) = \varepsilon_k \psi_k(\boldsymbol{r}_i)$$

Modern approaches use **preconditioning** (LOBPCG, HB+PP)[3,4] to damp high-energy modes that hinder convergence.

Convergence is studied using **dispersion**:

$$\langle \Delta h^2 \rangle^{(i)} = \frac{1}{A} \sum_{\ell=1}^{\Omega} \rho_{\ell\ell} \left[\langle \psi_\ell^{(i)} | h^2 | \psi_\ell^{(i)} \rangle - (\epsilon_\ell^{(i)})^2 \right]$$

Dispersion tells how far we are from the self-consistent solution. It is **independent** on the algorithm and **bound** by the chosen mesh.



Average dispersion of the eigenvalues in a ^{44}Ca calculation, including pairing correlations.

GCG **removes** the need for a preconditioner and **outperforms** currently available methods. Convergence **times** are reduced by ~40% with respect to LOBPCG.

3 APPLICATIONS: CLUSTERING, EXOTIC SHAPES, SUPERHEAVY NUCLEI

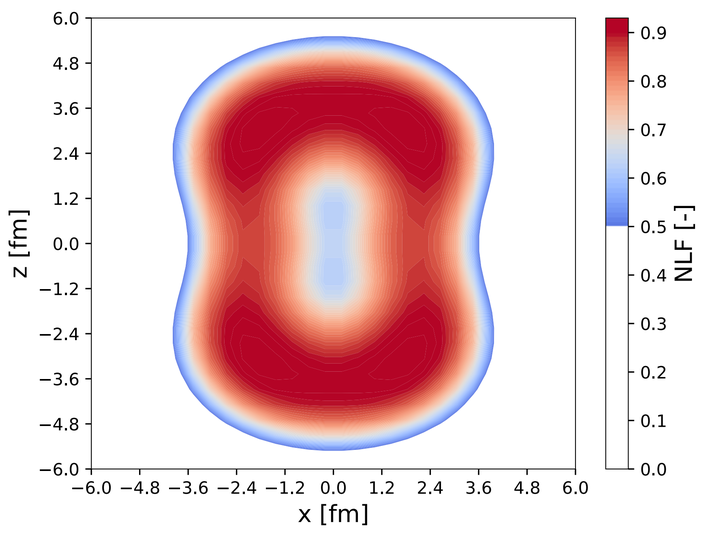
Using Cartesian meshes **new physics** becomes available.
Since **arbitrary** shapes can be represented, we can investigate **shape coexistance**, **symmetry breaking** and more.

Nuclear Clustering

High momenta can be easily reached compared to basis methods. Localized phenomena like **clustering** can be studied.

$$p_{\max} \sim \frac{\hbar}{\Delta x}$$

Nucleon Localization Function (NLF): given a particle in space, it tells the **probability** of finding another one in the near vicinity.



NLF in ^{24}Mg, highlighting clusters formation.

Pure Octupole shapes

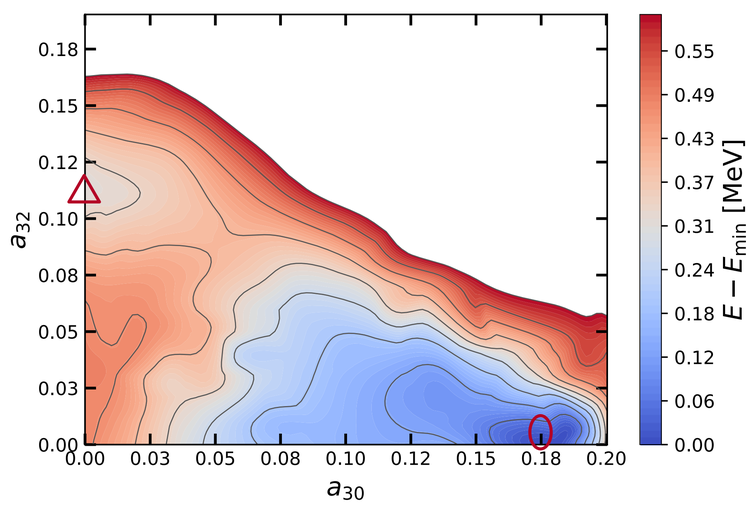
The increased **efficiency** of solution methods facilitates systematic studies of nuclear **deformations**.

Predictions made by macroscopic models and experiments can be studied in a **microscopic** fashion, such in the case of ^{110}Zr, where pure octupole configurations are predicted to be **lower** in energy[5].

By **constraining** momenta of the particle **density**, the energy surface can be explored.

$$E \mapsto E - \lambda \langle \hat{Q}_{3m} \rangle$$

Local minima in the surface may hint at possible shape coexistance. We find **two** local minima in ^{110}Zr corresponding to an **axial** octupole deformation and a purely **tetrahedrical** one.

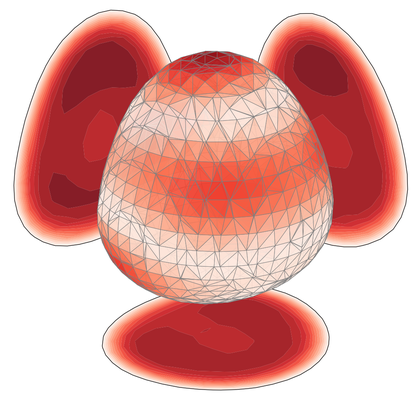


△ E: -893.1 MeV ○ E: -893.4 MeV

Superheavy elements

The **numerical** error of a mesh is primarily determined by the step size. Once fixed, the required CPU time scales **linearly** with mass, allowing systematic studies of superheavy **chains**.

Preliminary results can be obtained by minimizing the energy functional with selected self-consistent symmetries. Parity breaking calculations of 286No lower the energy of the spherical configuration.



^{286}No octupole deformed ground state. CPU time: ~3 hours

4 SUMMARY

The GCG method is successfully applied to reduce the **CPU** time of mesh Skyrme **energy density functional** calculations.

Fully **unconstrained** Cartesian meshes provide a promising tool for studying exotic shapes and super-heavy nuclei.

Future work will be focused on **systematic** studies at the extremes of the nuclide chart.

REFERENCES

[1] Colò (2020) Adv. Phys. X 5:1740061

[2] Li et al. (2020) CCF Trans. HPC 2:111–122

[3] Li, B., Ren, Z.-X., Zhao, P.-W. (2020) Phys. Rev. C 102, 044307

[4] Ryssens et al. (2019) EPJA 55:66

[5] F. F. Xu et al. (2024) Phys. Rev. C 109, 014311